Fahad Bin Islam

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Education

Long Island University (POST) | MS in Data Analytics and Strategic Business Intelligence | (GPA: 4.00/4.00) **Sept 2023 –May 2025**

- **Nespola Scholar Award – Recognized for academic excellence (2024–2025)**

Coursework: Statistics for Management, Data Visualization, Data-Driven Decision Making, Big Data & Machine Learning, Deep Learning, Generative AI

Professional Experience

Captor USA Corporation | Data Analyst – Healthcare Analytics **May 2024 - Present**

- Engineered SQL pipelines and automated ETL workflows using Python and Pandas, improving data processing speed by 35%.
- Designed and deployed interactive Tableau dashboards for payment analytics, reducing reporting time by 60%.
- Developed a secure Streamlit-based document upload app, enabling 400+ patients to submit files with 95% compliance.
- Integrated Google Drive API and PostgreSQL backend to support real-time data syncing across departments.
- Implemented Python-based automation for claim validation, saving ~20 hours/week in manual processing.
- Contributed to a targeted patient outreach campaign using segmentation models in scikit-learn, increasing engagement by 18%

Eastern Bank Ltd. | Data Analyst – MIS & Portfolio Analytics **Oct 2022 - Aug 2023**

- Developed automated EMI reporting pipelines using SQL and Python, reducing monthly reporting time by 50%.
- Created real-time dashboards in Power BI/Tableau for stakeholders like VISA, Mastercard and Bangladesh Bank.
- Led migration of portfolio data to a centralized warehouse using PostgreSQL, increasing data accuracy by 30%.
- Delivered weekly business review packs using Python, Power Query, and Excel, accelerating decision-making processes.
- Coordinated cross-functional data projects with finance and compliance teams to improve audit readiness and regulatory reporting.

The City Bank Ltd. | Data Analyst – MIS & Portfolio Analytics **Jul 2020 - Sep 2022**

- Automated daily transaction volume reporting using SQL and Excel, reducing manual workload by 40%.
- Developed a Random Forest model in Python to identify potential defaulters, improving early intervention by the collections team.
- Integrated time-series anomaly detection using Facebook Prophet to flag irregular spending behavior in retail credit data.
- Built Power BI dashboards for AMEX, VISA, and internal audit teams, streamlining compliance and executive reporting.
- Led a Python-based pipeline for portfolio risk segmentation using customer behavior clustering, supporting targeted engagement.

Advanced Chemical Industries Ltd. (ACI) | Planning Analyst – Conglomerate Analytics **May 2019 – Jul 2022**

- Led IoT and AI initiatives within ACI's Industry 4.0 transformation, optimizing supply chain operations using sensor-driven analytics.
- Built predictive inventory models in Python (Random Forest, LSTM) to reduce stockouts by 22% across key distribution zones.
- Automated daily sales tracking and forecasting using Power BI and Excel VBA, cutting reporting lag by 70%.
- Created interactive ecosystem visualizations in Tableau and PowerPoint to support strategic roadmapping and investor presentations.
- Partnered with engineering teams to analyze manufacturing telemetry data using R, identifying bottlenecks and reducing downtime by 15%.

Projects & Research Experience

Corona Product Assistant (RAG Chatbot) | Capstone Project | LIU Post **Jan 2025 - May 2025**

- Developed an intelligent Q&A chatbot using Retrieval-Augmented Generation (RAG) for Corona toilet models (Nyren, Aluvia, Cima).
- Engineered a dual-pipeline architecture with FAISS indexing, SentenceTransformer (text), and CLIP + GPT-4 Vision (image search).
- Built real-time memory tracking and feedback loops using Streamlit and st.session_state for coherent multi-turn conversations.
- Enabled dynamic document ingestion and tagging (PDF/DOCX) with semantic search—supporting uploads without restart.
- Integrated multi-agent capabilities using LangGraph, CrewAI, and AutoGen for document, query, and memory management.

Object Detection for Toilet Models | Deep Learning Project (YOLOv5) | LIU Post **Jan 2025 - May 2025**

- Built an object detection pipeline using YOLOv5 to classify Corona toilet models (Aquapro, Smart, Montecarlo) from product images.
- Annotated 500+ images using LabelImg and Roboflow, applying bounding box techniques and YOLO format structuring.
- Trained YOLOv5 on custom datasets with split folders (train/test/val), achieving 91% precision and 88% recall on test images.
- Evaluated performance using confusion matrices and IoU metrics; fine-tuned model using hyperparameter sweeps.
- Demonstrated results via a real-time inference GUI with bounding box overlays and class confidence scores.

Economic Order Forecasting | Advanced ML Project (Capstone) | LIU Post **Jan 2024 - May 2024**

- Forecasted order volume using SARIMAX, Random Forest, SVM, and PatchTST on 15 economic indicators.
- Engineered lag variables and expanded dataset from 17 to 182 features; used KNN imputation for missing value treatment.
- Achieved 88% R^2 with PatchTST and 0.34 MAE, outperforming traditional models in seasonal and long-term trend prediction.
- Used LASSO and Random Forest to rank key drivers like CPI, GDP, and unemployment.
- Conducted visual analysis of market cycles and downturn risks using time series peaks, troughs, and residual diagnostics.
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Crust & Craft Market Strategy | Applied Business Analytics Project | LIU Post **Sep 2024 - Dec 2024**

- Built an end-to-end strategy for a U.S. pizza startup using web scraping, trend mining, and predictive analytics.
- Scraped and analyzed 3,000+ Yelp reviews and Google Trends data using Python, PyTrends, and BeautifulSoup.
- Applied Random Forest to predict business success factors and Lasso Regression to optimize pricing strategy.
- Deployed LightGBM to forecast monthly optimal selling prices with 0.99 R^2 and 0.01 MAE.
- Visualized market saturation, state-level pricing trends, and demand peaks to guide location and menu decisions.

Hotel Booking Cancellation Prediction | Personal ML Project (Published on Medium) **May 2025 - Jun 2025**

- Built predictive models to classify hotel booking cancellations using Python and scikit-learn.
- Engineered features from booking data and handled class imbalance with SMOTE.
- Compared Logistic Regression, Random Forest, and XGBoost models using accuracy, ROC-AUC, and F1 score.
- Achieved 89% accuracy and 0.91 ROC-AUC using tuned Random Forest on 30k+ bookings.

Retail Discount Strategy Optimization | Data Visualization Project (Tableau) **Sep 2023 – Dec 2023**

- Designed 10 interactive dashboards in Tableau to analyze Superstore's discount, sales, and profit dynamics across regions and segments.
- Conducted "What-If Discount %" simulations to predict profitability impacts of varying discount rates by region and category.
- Identified underperforming categories (e.g., Furniture) and regions (Central, South) and recommended adjusted discount thresholds.
- Used KPI tracking, moving averages, and manufacturer-level profit breakdowns to drive actionable pricing insights.
- Recommended strategic timing of promotions and supplier optimization to boost quarterly profit margins.

Sales Analysis Dashboard | BI Reporting Project (Power BI) **Sep 2023 – Dec 2023**

- Developed a dynamic sales performance dashboard in Power BI using DAX and Power Query for end-to-end data transformation.
- Tracked KPIs such as Total Sales, Profit, Quantity Sold, and Discount Impact across product categories and timeframes.
- Created drill-down visuals and slicers to analyze regional performance and product-level profitability trends.
- Enabled executive-level reporting with dynamic filtering and trendline visualizations to support strategic planning.
- Enhanced user experience through interactive tooltips, bookmarks, and mobile-optimized layout.

Adaptive Bayesian Perception for 3D Object Recognition | Research Paper (In Progress) **Jan 2025 – Present**

- Proposed a Bayesian-driven framework for 3D object recognition in dynamic and occluded environments.
- Integrated feature selection modules combining keypoints, contours, and chromatic descriptors for probabilistic pose estimation.
- Designed a Monte Carlo sampling-based object tracking model using adaptive viewpoint refinement and evidence accumulation.
- Demonstrated improved recognition robustness in robotic navigation scenarios using stereo vision and contextual filters.
- Draft completed; submission targeted to Journal of Intelligent Robotic Systems in Fall 2025.

Skills

- **Languages:** Python, SQL, R, DAX, NoSQL | Familiar: Java, HTML/CSS, VBA
- **Libraries & Tools:** Scikit-Learn, TensorFlow, PyTorch, Pandas, NumPy, Matplotlib, Seaborn, XGBoost, LightGBM, StatsModels, SMOTE ,Power BI, Tableau, Excel (Pivot, VBA), Streamlit, BeautifulSoup, Selenium, FAISS, CLIP, SentenceTransformer
- **Frameworks & Platforms:** AWS (S3, EC2), Google Cloud (GCS, BigQuery), Flask, LangChain, LangGraph, CrewAI, FastAPI | Familiar: Hadoop, Snowflake
- **Database & Query Tools:** PostgreSQL, MySQL, Microsoft SQL Server, Google Big Query
- **Concepts:** RAG (Retrieval-Augmented Generation), Deep Learning, Generative AI, Predictive Modeling, Time Series Forecasting, A/B Testing, LLMs, Multi-Agent Systems, NLP, Bayesian Modeling, Statistical Inference, ETL Pipelines, CI/CD, Data Cleaning & Preprocessing.

Certification

- Generative AI and LLMs: Architecture and Data Preparation – IBM (Coursera)
- Advanced Relational Database and SQL – Coursera Project Network
- Data Manipulation with pandas – DataCamp
- Introduction to Statistics in Python – DataCamp